

Bhaskar Saripella

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Summary: Full Stack Software Developer with 6+ years of experience in application development and Machine Learning and AI across healthcare, finance, and FMCG domains. Proficient in Python, Java, and AWS, with expertise in Test Driven Development, Microservices, and Cloud computing. Also, I have experience managing risk ensuring compliance with both operational and regulatory risk management frameworks, including adherence to US FDA 21 CFR 820 for quality system regulations in medical device and healthcare-related industries.

PROFESSIONAL EXPERIENCE:

Machine Learning Engineer (Freelance) | AppTechno Vision | Toronto, Canada Sep 2024 – Present

- Addressed over 10 critical security vulnerabilities and enhanced AI model training.
- Developed prompts, test cases and corrective responses to resolve over 50 model failures.
- Generated a Generative AI concept that integrates Computer Vision to help customers find products in a store.
- Collaborated with stakeholders to align AI/ML solutions with business and technical objectives.
- Designed dynamic web application dashboards using AngularJS and React with NodeJS backend.

Lead Engineer | HCL Technologies | Hyderabad, India June 2022 – July 2024

- Developed a Generative AI concept to identify alternative drugs and off-label drug combinations for a given illness.
- Resolved over 200 static code and open-source software vulnerabilities within the first 30 days of employment.
- Integrated data lakes and warehouses using HDFS, S3, and MySQL for clinical data and SAS analytics.
- Performed preventative maintenance, proactive fault rectifications to ensure ZERO defect production deployment.
- Migrated core technologies (Spring, Hibernate, AWS SDK) and upgraded system performance.
- Developed the SpringBoot Java backend for a credit card fraud detection system, improving fraud capture by 30%.
- Led cloud infrastructure optimization on AWS (Lambda, S3, load balancers), reducing costs by 15%.
- Managed Proxies, DNS, Load-Balancers and testing of connectivity between micro-services on AWS and on-prem.
- Managed and developed retirement procedures for applications hosted on AWS Infrastructure.

Data Scientist | 3Frames Software Lab | Bangalore, India Aug 2018 – Jan 2022

- Generated UI models to integrate frontend APIs with REST, GRPC and SOAP APIs from Java and Python backends.
- Integrated machine learning models into existing applications, resulting in \$300,000 in sales.
- Designed a campaign generator using collaborative filtering algorithms, improving customer engagement by 10%.
- Developed a predictive models using probabilistic and regression algorithms to enhance sales revenue by 30%.
- Developed and optimized over 20 complex SQL queries including stored procedures, and ETL processes in SQL Server, MySQL and Oracle SQL, data warehouses (BigQuery, Snowflake) and MongoDB NoSQL databases.
- Deployed Machine Learning modules on AWS EC2 instances either as monolith or microservices.

Development Engineer | Zimmer Biomet | Warsaw, IN USA Jun 2016 – July 2017

- Developed a Computer Vision module to detect wear and tear on returned instruments, improving condition assessment accuracy and operational efficiency.
- Fine-tuned a pre-trained machine learning model using over 1,000 annotated images to classify and catalog 11,000 instruments and implants
- Led data-driven validation of legacy Zimmer instruments for Lifespan Legacy field use, supporting design control and regulatory compliance as per 21 CFR 820.
- Authored 4 comprehensive research protocols (ZTP, ZRM, and ZRR) to evaluate and simulate field wear conditions.

Quality Engineer | Engineered Plastic Components | Rantoul, IL USA

Jun 2015 – Apr 2016

- Analyzed quality logs, metrics, and specifications to identify critical failure metrics and enhance Standard Operating Procedures (SOPs) across 54 product lines.
- Leveraged statistical analysis and Linear Programming algorithms to optimize machine uptime and minimize idle time, resulting in over 50 hours of improved operational efficiency.

Education

Master of Science in Mechanical Engineering | Missouri University of Science and Technology | Rolla, MO USA | 2015

Bachelor of Engineering in Mechanical Engineering | Osmania University | Hyderabad, India | 2012

Certifications

Data Engineer Associate, DataCamp Nov 2024

Intel Edge AI Certification, Intel May 2024

Fundamentals of Parallelism on Intel Architecture: authorized by Intel, Coursera Dec 2020

Deep Learning Specialization: Dr. Andrew Ng by Deeplearning.ai (Stanford University), Feb 2019

Internships

Intern | Indian Institute of Technology Hyderabad, India

Feb 2022 – June 2022

- Applied Generative Adversarial Neural Networks in robotic stability
- Studied applications of Generative Functions and in Generative AI LLM algorithms
- Feature engineering for Day Trading systems using time series decomposition techniques

Intern | Indian Institute of Technology Hyderabad, India

Apr 2018 – Aug 2018

- Applications of optimization algorithms to optimize cost function convergence using Newtonian and Stochastic gradient descent techniques
- Extended Neural Networks in Machine Learning and explored industry applications in India

Skills:

Languages & Frameworks: Java, Python, Spring, Hibernate, Node.js, .NET, RESTful APIs, Microservices, Kafka, PL/SQL

Cloud Platforms: AWS (Lambda, S3, EC2, ECS, Fargate), Cloud Architecture, Serverless Computing, CI/CD

Databases & Data Integration: MySQL, Oracle, Postgres, HDFS, Data Warehouses, Clinical Data Lakes

DevOps & Version Control: Git, Ansible, Docker, Jenkins

Tools & Platforms: Swagger, Postman, RedHat Linux, WebSphere, Tomcat, Redis, XML/JSON, MQ

AI/ML: Scikit-Learn, Pytorch, Tensorflow, AutoML